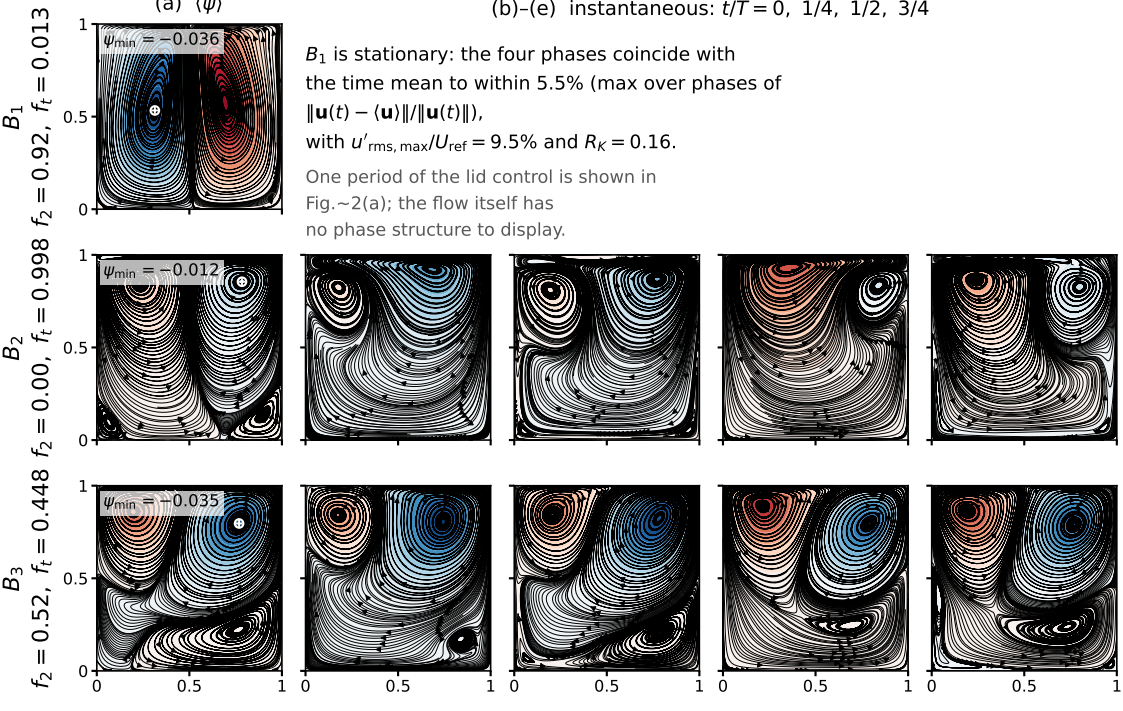

(a) ⟨ψ⟩
(b)–(e) instantaneous: t/T = 0, 1/4, 1/2, 3/4
B1 f2 = 0.92, ft = 0.013
B2 f2 = 0.00, ft = 0.998
B3 f2 = 0.52, ft = 0.448
ψmin = −0.036
ψmin = −0.012
ψmin = −0.035
0
0.5
1
B1 is stationary: the four phases coincide with
the time mean to within 5.5% (max over phases of
‖u(t) − ⟨u⟩‖/‖u(t)‖),
with u′rms, max/Uref = 9.5% and RK = 0.16.
One period of the lid control is shown in
Fig.~2(a); the flow itself has
no phase structure to display.
one scale and one level set for all eleven field panels
streamlines: same density and placement rule
−2.10
0
2.10
stream function ψ/(Uref) [×10−2]